

MIGRATION

A Research Synopsis

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A NOTE ON METHOD

Each section of “Migration” began with a bird. I researched the documented biology and migratory behavior of various species, taking particular note of elements like geographic migrational route, its triggers for migration, its vulnerabilities on the journey, the ecological pressures involved. From there I identified human migration stories that seemed to move along similar channels of danger, instinct, seasonality, survival, displacement, and return. Rather than being academic, the human sources that informed my writing were largely first-person narrative, immersive journalism memoirs, and oral testimony, as these kinds of accounts are capable of preserving not only factual information but emotional and lived experience surrounding migration itself. From this research, I developed a series of creative/poetic pieces that weave the parallel lines between human and avian migration, each migratory example having a specific, corresponding parallel.

01 | ONE JOURNEY, MANY WINGS

Bird: Sandhill Crane — Platte River Flyway

Every spring, roughly 600,000 to one million Sandhill cranes congregate for weeks along the Central Platte River Valley in Nebraska where they fuel up on waste grain before continuing on their journey north. Within these immense migratory flocks, each individual bird is steadied by the wingbeats of others — This massive group migratory event is documented across ornithological literature and crane tracking projects, including ongoing work by the Audubon Society.

Human Migration: The Hajj Pilgrimage

The parallel drawn in this section is annual Hajj pilgrimage in which millions of muslims move shoulder to shoulder on one of the oldest repeated migrations in human history, grounded in first-person accounts of the Hajj including historical Islamic scholarship tracing the roots of the pilgrimage through the story of Ibrahim at least fourteen centuries back.

Audubon — “Tracking Data Shows Us One of North America’s Most Epic Migrations” (2024) — *Overview of Sandhill crane migration patterns across the Platte River flyway using modern tracking data, emphasizing the immense scale, coordination, and endurance of one of North America’s oldest migratory routes.*

Detroit Mom — “Journey of a Lifetime: My Teenage Hajj Story” (2022) — *A woman’s personal account of performing Hajj as a teenager, describing the various holy sites along the pilgrimage and the experience of being swept into a massive human current moving as one body.*

Richard Francis Burton — Personal Narrative of a Pilgrimage to Al-Madinah & Meccah (1855) — *A foundational first-person narrative of the Hajj pilgrimage, notable for its granular detail of the pilgrimage as a physical and spiritual migration — one of the earliest Western accounts of the collective movement of pilgrims toward Mina.*

02 | THE DISTANCE OF ONE HEART

Bird: Ruby-Throated Hummingbird — Crossing of the Gulf of Mexico

The ruby-throated hummingbird weighs approximately three grams and flies across 500 hazardous miles of open water for roughly 18–22 hours without stopping once. The image of a "three-gram jeweled needle" threading itself through a dangerous landscape draws directly from the documented biology of this dangerous crossing in ornithological research and GIS data.

Human Migration: Asylum Seekers Crossing the Gulf Coast Corridor

The human parallel in this section is drawn from the documented experiences of Central American and Caribbean asylum seekers navigating the Gulf region on foot and across water, for whom stopping is equally unsurvivable.

Sheri Williamson — “The Fall Migration of Ruby-Throated Hummingbirds” (2020) — *Scientific narrative analysis of ruby-throated hummingbird Gulf migration, including detailed description of what the crossing looks and feels like from the data: departure at dusk, 500 miles of dark water, arrival trembling and depleted.*

03 | FOLLOWING THE STARS

Bird: Indigo Bunting — Stellar Navigation via Polaris

The indigo bunting is one of the most masterful celestial navigators in the avian world. Research by the Cornell Lab of Ornithology and others has established that buntings imprint on the celestial north pole during a developmental window and use this fixed point to orient their migration. The bunting moves at night, when hawks cannot trace its path and when the world falls quiet enough for the journey to be survivable.

Human Migration: The Underground Railroad

The parallel drawn here is that Polaris guided both migrating buntings and enslaved Africans fleeing to their freedom on the Underground Railroad. Abolitionist records and preserved oral histories refer to the night, the use of stars, and the need for silence among predators that made daylight movement fatal, all conditions which mirror that of the bunting.

Stephen Emlen — “The Stellar-Orientation System of a Migratory Bird” (1975) — *An ornithological study on celestial navigation in indigo buntings, demonstrating how they orient themselves using the rotational center of the night sky during a critical developmental period.*

Chanda Prescod-Weinstein, Ms. Magazine — “Harriet Tubman, Astronomer Extraordinaire” (2022) — *A contemporary essay weaving slave narrative testimony with astronomical science to argue for the literal, technical use of Polaris in Black liberation.*

North Country Underground Railroad Historical Association — “Our Logo” (n.d.) — *An historical explanation of the North Star’s symbolism within Underground Railroad memory and iconography, reinforcing the role of Polaris as both a literal navigational tool and enduring symbol of movement toward freedom.*

04 | THE WEIGHT OF AN EARLY JOURNEY

Bird: Caspian Tern — Chick Migration Without Prior Knowledge

Research on the caspian tern documents that chicks begin migration before they have any independent knowledge of the route, following paternal guidance for survival along the way. If the guiding father is killed mid-route (particularly by colliding with fog-obscured environmental hazards, a documented cause of tern mortality), then the chick is left to navigate alone without a map, a compass, or the knowledge it needs to survive.

Human Migration: Children on La Bestia

This section traces the parallel between the orphaned tern chick and the thousands of unaccompanied children who ride the Mexican freight train, *La Bestia* (The Beast), to the northern border. Children as young as ten are forced to improvise this journey alone after having lost their guiding parent to one of the frequently occurring tragedies of injury, death, or mid-journey deportation on *La Bestia*.

Patrik Byholm, et al. — “Paternal transmission of migration knowledge in a long-distance bird migrant” (2022) — *GPS-tracking study demonstrating that young Caspian terns rely on paternal guidance during their first journey south, establishing migration as learned, relational knowledge rather than pure instinct in this species.*

Sarah Yolanda McClure, Berkeley— “Migration: A Central American Odyssey” (2013) — *Documents Nazario's methodology, including the moment she nearly fell from the train after being struck by an overhanging branch, the same sort of invisible structure that suddenly kills Caspian terns in fog.*

05 | MOVED BY THE EARTH

Bird: Snowy Owl — Climate Displacement

When lemming populations collapse due to Arctic ice loss and permafrost thaw, the snowy owl is pushed southward into unfamiliar landscapes. Its white feathers, evolved over millennia for the snowy tundra, make the owl immediately visible against the warmer southern landscapes that cannot hide it.

Human Migration: Climate Displacement (Sahel)

The snowy owl parallel draws on the human displacement of the Great Sahelian Drought, with widespread famine and over 100,000 deaths during its most devastating period from the late 1960s through 80s. Periodic droughts still occur in 2026 as the region warms 1.5 times faster than the global average, and to this day, failed rains and loss of topsoil force families to start from the ground up in new cities, economies, and ways of life.

Denver Holt, et al. — “A Worldwide Review of Snowy Owl Feeding Ecology” (2024) — *Comprehensive review of snowy owl feeding ecology emphasizing the species heavy reliance on cyclic lemming and vole populations.*

Climate and Migration Coalition — “Moving Stories: The Sahel” (2013)— *First-person and community-centered accounts of climate displacement across the Sahel, documenting how drought, failed harvest, and environmental instability reshape migration decisions, family structures, and everyday survival.*

06 | WHERE RECOVERY BEGINS

Bird: Painted Bunting — Dangerous Molt

From late summer through early fall each year, painted buntings undergo a “molt migration”, rare for songbirds, in which they travel to northern Mexico to molt before continuing their journey south. Ornithological research documents that during this molt, the bunting’s bright feathers are stripped away and it must hunker down for weeks, exposed and unable to fly freely, while they slowly restore their full, vibrant feathers.

Human Migration: Medical Migration

The painted bunting parallel draws on the experience of those who migrate specifically to receive medical treatment during which they are often stripped of their normal environment, exposed and dependent, and navigating in a place where they may not speak the language, all the while a physical crisis is occurring beneath the surface. This experience is often termed “medical tourism”, and it is candidly revealed through first-person accounts of the search for life-saving treatment abroad.

Christopher Thompson — “The Sequence of Molts and Plumages in Painted Buntings and implications for Theories of Delayed Plumage Maturation” (1991) — *Detailed ornithological study of painted bunting molt cycles and plumage transition, documenting the biologically vulnerable period during which buntings temporarily lose the bright feathering that normally protects and defines them.*

Lewis Fein, Medical Tourism Magazine — “A Tourist of Hope: My Story About Medical Tourism” (2026) — *A first-person account of traveling abroad for life-saving treatment, the experience of medical necessity as a form of forced migration.*

07 | FLYING TOWARD THE HARVEST

Bird: Fork-Tailed Flycatcher — Following Insect Abundance

The fork-tailed flycatcher migrates seasonally from South to North America, following high-precipitation areas for the attendant insect abundance therein. Avian ecological research on foraging behavior reveals that this flycatcher is a nomadic species with a migratory route shaped not by a fixed destination or a fixed time, but by the inter-annually shifting condition of resource availability.

Human Migration: Migrant Agricultural Workers Following the Harvest

The human parallel of a man in the back of a truck whose intuition leads him to the harvest and who can read whether the rows will be good in a particular week is drawn from documented accounts of migrant agricultural workers who move through the US in circuits as calculated as any avian migration route, following the apple harvest, the onion harvest, the citrus. Their embodied knowledge of crops mirrors the flycatcher's embodied knowledge of insects.

Keith Hobson, et al. — “A trophic niche shift in a South American migrant: Stable nitrogen isotope signatures in feathers of Fork-tailed Flycatchers (*Tyrannus savana*)” (2020) — *Stable isotope analysis tracing dietary and foraging shifts in fork-tailed flycatchers across their migratory route, supporting the idea that their movement patterns are closely tied to changing resource availability and seasonal insect abundance.*

Natascha Klocker et al., Agriculture and Human Values — “Exploring migrants’ knowledge and skill in seasonal farm work: more than labouring bodies” (2020) — *Research emphasizing the highly specialized environmental knowledge held by migrant agricultural workers, particularly their embodied understandings of seasonal cycles, crops, weather, and harvest timing across shifting regions.*

Dan Charles, NPR — “Inside The Lives Of Farmworkers: Top 5 Lessons I Learned On The Ground” (2016) — *First-person field journalism from apple orchards, blueberry fields, and strawberry farms. Workers describe the circuit as a way of life, the knowledge worn into the body season after season.*

08 | LEARNING ALONG THE WAY

Bird: Whooping Crane — Recovering Through Learned Migration

By the 1940s, there were only sixteen whooping cranes left in the world, and their flyway knowledge, passed from parent to chick over generations, had nearly been extinguished. Conservation biologists developed a method demonstrating survival to the chicks, involving handlers in white crane costumes guiding natural behaviors and ultralight aircrafts leading the birds south along the Eastern Flyway. With heavy involvement by organizations like the International Crane Foundation, this story is one of the most documented inter-species acts of knowledge transfer in conservation history.

Human Migration: Immigrant Mutual Aid Networks

The human parallel is the mutual aid network made manifest in WhatsApp chains that warn of checkpoints, in congregations that keep open cots, in decades-settled families who pass along the names of clinics that ask no questions. Survival knowledge for newcomers is distributed the same way the crane's flyway was re-taught: not from a map, but from someone who has already survived the route.

International Crane Foundation — “Connecting with Whooping Cranes” (n.d.) — *Overview of whooping crane migration, conservation, and recovery efforts, including flyway learning, reintroduction programs, and the fragile transfer of migratory knowledge between generations.*

Amanda Michelle Gomez, DCist — “Inside the Local Mutual Aid Effort Supporting the Migrants Texas Bused to D.C.” (2022) — *Field and interview-based account documenting the volunteer mutual-aid networks that mobilized to support migrants arriving in Washington, D.C. after being bused from Texas, ultimately depicting survival knowledge passed through churches, WhatsApp chains, temporary housing networks, and community organizers.*

09 | BETWEEN SHORE AND HORIZON

Bird: Wandering Albatross — A Natural Pull to Wander the Ocean

With the largest wingspan, up to 11 feet) of any living bird, the wandering albatross spends the majority of its life in uninterrupted flight across the Southern Ocean, often with no fixed destination, and touching land only briefly to breed. It utilizes a specialized shoulder-locking mechanism and “dynamic soaring” soar up to 650 miles in a single stretch without flapping its wings once. The clearest truth gleaned from the documented biology of this fascinating engineering is that the albatross does not wander because it is lost, but because wandering is its nature.

Human Migration: Wanderlust

The human parallel in this section is the pull toward movement that exists independent of political or economic necessity, drawing on the Aeon essay “A Dispatch from the End of Travel’s Brief, Troubled Golden Age,” whose author distinguishes between *wanderlust* and the more precise term *dromomania*, the latter capturing the obsessive, involuntary quality of the drive to move; not a preference, but a condition, almost as if second nature.

Paul Clammer, Swoop — “Following the Albatross, Sentinels of Antarctica” (2024) — *Narrative natural history account of the wandering albatross, emphasizing its immense wingspan, oceanic lifestyle, and near-perpetual flight across the Southern Ocean.*

Aeon — “A Dispatch from the End of Travel’s Brief, Troubled Golden Age” (2020) — *First-person reflective essay examining compulsive movement, travel, and the psychology of wanderlust in the modern era.*

10 | CARRYING HOME ACROSS DISTANCE

Bird: Cattle Egret — Natural Trans-Atlantic Dispersal

The cattle egret emigrated naturally from Africa to the Americas, its first sighting in the Caribbean being recorded in 1877. Since then, it has spread across the continent, and while its behavior gradually molds to the new environment, its amber and white plumage, across every range expansion, has remained unchanged.

Human Migration: Caribbean Diaspora

This section draws on my family’s migration from Trinidad to the US. The egret’s parallel to Caribbean diaspora experience became clear to me after a conversation that I recently had with my dad, in which he noted that the Trini accent, its particular music and rhythm, functions as memory itself. Even when it fades on the tongue after years away from the island, the ability to hear it in one’s heart is the voice refusing erasure and remaining unchanged.

Justine Hausheer, The Nature Conservancy — “How Cattle Egrets Took Over the World” (2021) — *A documentation of how the cattle egret achieved a natural, trans-Atlantic dispersal from Africa into the Americas, documenting how the species adapted to new environments across its newly-charted range.*

Closing Research Note

Though this project draws parallels between avian and human migration, it also provides an avenue through which to recognize the profound asymmetry between them. Birds move through the world according to season, instinct, hunger, reproduction, and the shifting availability of resources, crossing continents through skies untouched by the borders below. Human movement, by contrast, is layered with politics, survival, memory, faith, labor, displacement, and longing. We migrate to flee enslavement and instability, to seek safety and work, to recover from illness, to endure environmental collapse, to honor devotion, to pursue the possibility of another life beyond the horizon. Birds and humans both migrate, but only one encounters invisible structures and checkpoints along the way.

PRIMARY SOURCES & SELECTED BIBLIOGRAPHY

Narrative Accounts & Memoir

Burton, Richard Francis. *Personal Narrative of a Pilgrimage to Al-Madinah & Meccah*. London: Longman, Brown, Green, and Longmans, 1855.

Detroit Mom. "The Journey of a Lifetime: My Teenage Hajj Story." *Detroit Mom*, 7 July 2022.

Medical Tourism Magazine. "A Tourist of Hope: My Story About Medical Tourism." *Medical Tourism Magazine*. 2026

Aeon. "A Dispatch from the End of Travel's Brief, Troubled Golden Age." *Aeon Essays*, 2020.

Immersive Journalism & Oral Testimony

Charles, Dan. "Inside the Lives of Farmworkers." *NPR: The Salt*, July 2016.

Gomez, Amanda Michelle . "Inside the Local Mutual Aid Effort Supporting the Migrants Texas Bused to D.C." *DCist / WAMU*, 24 May 2022.

Yolanda, Sarah MacClure. Center for Latin American & Caribbean Studies, Berkeley. "Migration: A Central American Odyssey." University of California, Berkeley, 2014.

Climate and Migration Coalition. "Moving Stories: The Sahel." *Climate Migration Coalition*. 2013

Ornithological & Scientific Sources

National Audubon Society. "Tracking Data Show Us One of North America's Most Epic Migrations." *Audubon*, 10 May 2024.

Williamson, Sheri. "The Fall Migration of Ruby-Throated Hummingbirds." *Field Guide to Hummingbirds*, 2020.

Emlen, Stephen T. "The Stellar-Orientation System of a Migratory Bird." *Scientific American*, vol. 233, no. 2, Aug. 1975, pp. 102–111.

Byholm, Patrik, et al. "Paternal Transmission of Migration Knowledge in a Long-Distance Bird Migrant." *Nature Communications*, vol. 13, 2022.

Holt, Denver W., Matthew D. Larson, Mathew T. Seidensticker, and Stephen P. Hiro. *A Worldwide Review of Snowy Owl Feeding Ecology: The Importance of Lemmings and Voles in a Changing Climate*. 2024

Thompson, Christopher W. "The Sequence of Molts and Plumages in Painted Buntings and Implications for Theories of Delayed Plumage Maturation." *The Condor*, vol. 93, no. 2, May 1991, pp. 209–235.

Hobson, Keith A., et al. "A Trophic Niche Shift in a South American Migrant: Stable Nitrogen Isotope Signatures in Feathers of Fork-Tailed Flycatchers (*Tyrannus savana*)."

International Crane Foundation. "Connect with Whooping Cranes." *International Crane Foundation*. n.d.

Clammer, Paul. "Following the Albatross, Sentinels of Antarctica." *Swoop Antarctica Blog*, 13 June 2024.

Hausheer, Justine. "How Cattle Egrets Took Over the World." *The Nature Conservancy Blog*, 26 April 2021.

Historical & Scholarly Sources

Prescod-Weinstein, Chanda. "Harriet Tubman, Astronomer Extraordinaire." *Ms. Magazine*, Feb. 2022.

North Country Underground Railroad Historical Association. "Our Logo." *NCUGRHA*, n.d.

Ananth, Natascha, et al. "Exploring Migrants' Knowledge and Skill in Seasonal Farm Work: More Than Labouring Bodies." *Agriculture and Human Values*, vol. 37, no. 2, 2020, pp. 463–478.